

WASP-12b

R-2272

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-12_150103 · created 2026-08-16T23:04:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457025.79157 +/- 0.00012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.098 +/- 0.012
Transit depth (Rp/Rs)^2	0.95 +/- 0.23 [%]
Orbital Inclination (inc)	81.75 +/- 2.22
Airmass coefficient 1 (a1)	0.9414 +/- 0.0071
Airmass coefficient 2 (a2)	0.055 +/- 0.022
Scatter in the residuals of the lightcurve fit is	0.76 %
Best Comparison Star	#2 - [389, 279]
Optimal Method	PSF photometry
Transit Duration (day)	0.1179 +/- 0.0059

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.098 ± 0.012 vs 0.1178 (deviation 0.96σ)
Duration fit vs published	0.1179 d vs 0.125 d (deviation 0.7σ)
Mid-time O–C	-0.3 min
Depth SNR	4.8
Residual scatter	0.76 %

Light curve

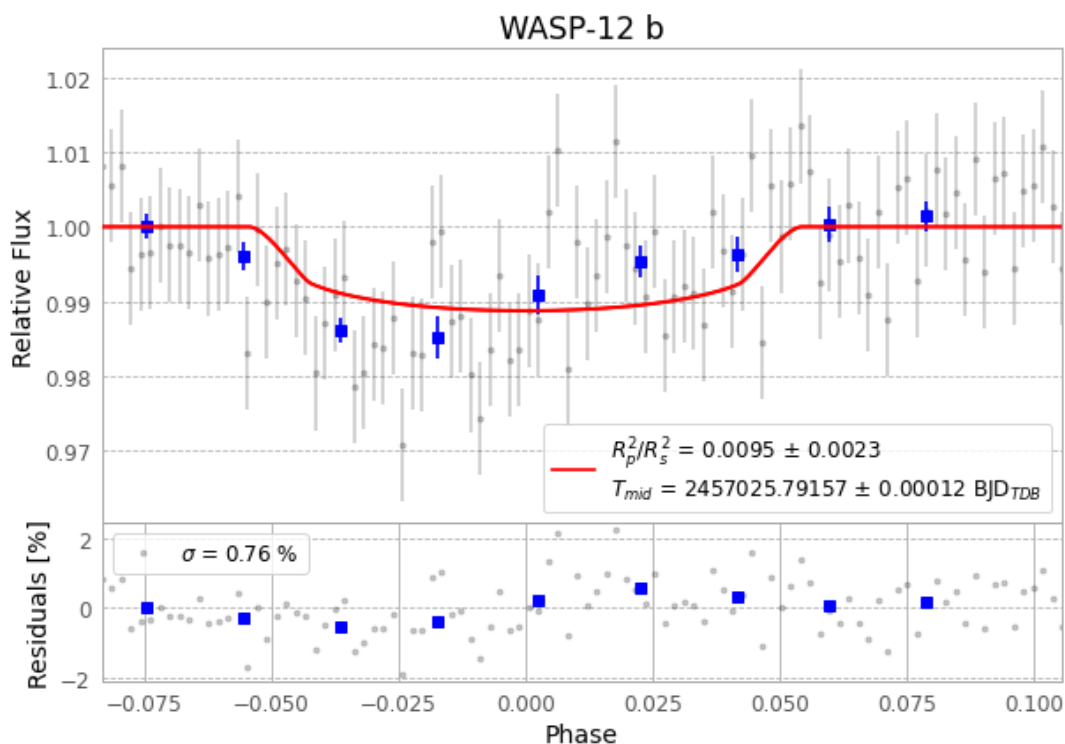


Figure 1 — FinalLightCurve_WASP-12 b_2015-01-02.png (SHA-256 57315df9266a0f40...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [407, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8dfabf23822271d6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

100 FITS frames (66 MB), WASP-12150103043912.FITS → WASP-12150103093612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-150103.log (6cf2fe9c036f...), FinalLightCurve_WASP-12 b_2015-01-02.png (57315df9266a...), FinalLightCurve_WASP-12 b_2015-01-02.csv (aab035100886...)

Compute resources

Wall time 153.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-16 23:04Z.